

User Experience and Performance During Immersive Avatar Customization

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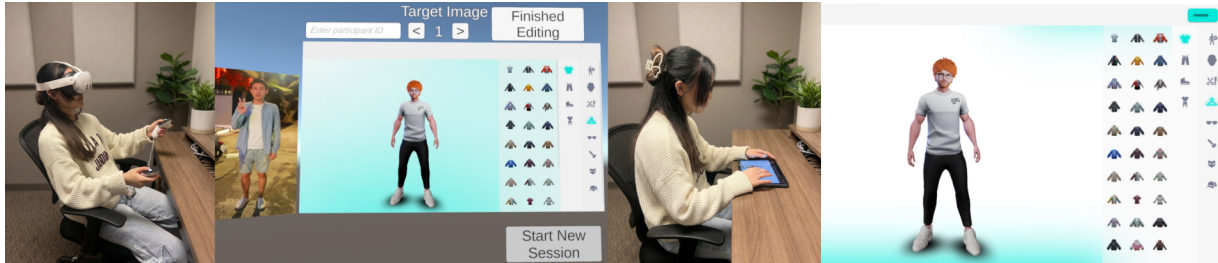


Figure 1: A user interacting with the avatar customization interface deployed in this study. The left two images depict a participant interacting with the VR interface and the right two images a participant using the tablet interface.

ABSTRACT

In this work, we present a methodology that treats avatar customization as a measurable cognitive-perceptual task and report results from a pilot study examining how presenting the same customization interface across different mediums affects user experience and performance. Participants completed an avatar target matching task on either a tablet or a VR headset. Interaction metrics, subjective workload and usability measures, and similarity ratings were analyzed. While subjective measures did not differ substantially, preliminary results showed VR users tended to perform fewer actions and take longer between actions. We discuss implications for future studies investigating immersive customization interface design.

Index Terms: avatar customization, virtual reality, usability, workload, pilot study

1 INTRODUCTION

Avatars, or virtual representations of users' bodies, are a fundamental component of immersive virtual reality (VR) environments. They convey information that supports spatial awareness, action control, and social interaction. Beyond providing functional information to users, their pliability allows users to embody variations in appearance that can meaningfully alter perceptions, behaviors, and interactions [3, 4]. As VR applications for social, gaming, and professional contexts grow, effective avatar customization interfaces become increasingly important. However, little is known about how immersive interface design shapes user experience and performance during customization.

We present a methodology that treats avatar customization as a measurable cognitive-perceptual task, enabling the study of how customization interface design affects user experience and performance. Current VR avatar customization interfaces vary widely in affordances and interaction styles. Many systems rely on traditional 2D menu panels for option selection (e.g., *Horizon Workrooms* and *Spatial*), while others employ more immersive approaches, such

as diegetic in-world tools that embed customization actions into the environment (e.g., *Rec Room*'s avatar costume editor). These interfaces also differ substantially in the granularity of available customization options and the fidelity of resulting avatars. To our knowledge, no standardized immersive customization interface exists, nor is there a clear understanding of how users experience these systems.

To test our methodology, we conducted a pilot examining how presenting the same customization interface across different mediums affects user experience and performance. We asked:

RQ1. How does medium affect cognitive load and perceived system usability during customization?

RQ2. How does medium affect customization behaviors and outcomes?

Initial results supported our methodology, showing that while subjective measures did not differ strongly, VR users trended toward performing fewer actions and taking longer between actions. We view these findings as an initial step toward understanding effective immersive customization interface design.

2 PILOT EXPERIMENT

2.1 Participants

Twenty participants ($M = 14$, $F = 4$, $TF = 2^1$; ages 20 – 34, $SD = 3.12$) completed the pilot. Participants were required to be at least 18 years old, have normal or corrected-to-normal vision, and no history of severe motion sickness. Each participant was compensated \$10/hour via Amazon gift card. The study protocol was approved by our institution's ethics board. Participants also reported their prior VR experience: 5.3% had never used VR, 26.3% had used it once or twice, 42.1% a few times per year, 10.5% a few times per month, and 15.8% a few times per week.

2.2 Materials and Measures

We deployed a custom implementation of the Ready Player Me (RPM) avatar customization interface² on an Apple iPad and a Meta Quest 3 using Unity 2021.3.37f1 (See Figure 1). On the iPad, RPM was displayed via web browser; on the Meta Quest 3, it was presented in a fixed webview³ positioned 1.436 m in front of the user.

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¹TF indicates transgender female.

²<https://readyplayer.me/>

³<https://github.com/TLabAltoh/TLabWebViewVR>

Interaction in VR was performed using raycast selection with hand-held controllers. On both devices, pressing the “finish” button directed participants to a survey in which they rated perceived success and difficulty in matching the target image.

Both interfaces logged session start and end times, number of edits, and a timestamped record of each action type and parameter value. Data were serialized as JSON and uploaded to a database, and a GLB file of the final avatar was generated via RPM.

Twelve target images (5 “easy”, 5 “hard”, 2 practice) for participants to match were selected from the SSHQ dataset⁴, whose license permits use for academic research purposes.

Participants completed the NASA-TLX [2] to assess subjective workload and the System Usability Scale (SUS) [1] to assess perceived usability.

2.3 Procedure

Our pilot employed a between-subjects design in which participants completed the experiment using either an iPad or a Meta Quest 3. After providing informed consent, participants were introduced to the avatar customization interface; those in the VR condition were additionally assisted with headset fit and interpupillary distance adjustment. All participants then completed a 10-minute training phase to practice the task. Following training, participants rated perceived success and difficulty and completed the NASA-TLX.

The experimental task consisted of two 10-minute avatar creation sessions. Each participant completed one “easy” and one “hard” target image, selected from the pilot image set and counter-balanced across participants. Participants were instructed to match each target as accurately as possible using the customization interface. After each session, participants again rated perceived success and task difficulty and completed the NASA-TLX. Finally, participants completed the SUS to assess overall interface usability. The total study duration was approximately 45 minutes.

3 RESULTS

Statistical analyses were conducted in R version 4.3.1. Data were inspected for outliers, and normality and homogeneity of variance were checked using Shapiro-Wilk and Levene’s tests.

Objective Measures. Objective measures extracted from our action logs included **time to completion**, **number of actions**, and **time between actions**, which characterize customization behavior and efficiency (RQ2) and provide indirect indicators of cognitive demand during interaction (RQ1).

An independent samples t-test showed no significant difference in time to completion between the tablet ($M = 428.84$ s, $SD = 95.77$) and VR ($M = 478.98$ s, $SD = 145.66$) conditions, $t(18) = -0.91$, $p = 0.38$, $d = -0.41$. Participants in the tablet condition ($M = 153.25$ actions, $SD = 30.04$) tended to perform more actions than those in VR ($M = 119.60$ actions, $SD = 42.52$), although the difference was not statistically significant, $t(18) = 2.04$, $p = 0.056$, $d = 0.91$. Finally, participants in VR had significantly longer time between actions ($M = 3.44$ s, $SD = 0.59$) compared to the tablet condition ($M = 2.76$ s, $SD = 0.29$), $t(18) = -3.22$, $p = 0.0047$, $d = -1.44$. Together, these results suggest that while overall task completion time was similar, the two mediums may support different patterns of interaction during customization (RQ2).

Similarity Ratings. Two independent raters assessed the **similarity of each final avatar to the target image** (RQ2) on a 1-7 Likert scale (1 = *not similar at all*, 7 = *extremely similar*). Inter-rater reliability was evaluated using a two-way random effects intraclass correlation coefficient (ICC[2,1]). Reliability was excellent (ICC=0.9), so the ratings were averaged for analysis. An independent samples t-test revealed no significant difference in similarity

ratings between avatars created in the tablet ($M = 4.33$, $SD = 0.76$) and VR ($M = 4.10$, $SD = 0.64$) conditions, $t(18) = 0.72$, $p = 0.481$, $d = 0.32$. This suggests that medium did not meaningfully affect final customization outcomes (RQ2).

Questionnaires. The NASA-TLX, perceived difficulty and success items, and SUS provided measures of subjective cognitive load and perceived system usability (RQ1).

Independent samples t-tests revealed no significant differences between the tablet and VR conditions on any measure: NASA-TLX (tablet $M = 2.53$, $SD = 0.78$; VR $M = 2.45$, $SD = 0.64$; $t(18) = 0.23$, $p = 0.82$, $d = 0.11$), perceived success (tablet $M = 4.05$, $SD = 0.76$; VR $M = 4.25$, $SD = 1.32$; $t(18) = -0.42$, $p = 0.68$, $d = -0.19$), perceived difficulty (tablet $M = 5.05$, $SD = 0.55$; VR $M = 4.75$, $SD = 1.06$; $t(18) = 0.79$, $p = 0.44$, $d = 0.36$), and SUS (tablet $M = 75.5$, $SD = 16.45$; VR $M = 73.5$, $SD = 9.73$; $t(18) = 0.33$, $p = 0.74$, $d = 0.15$). These results suggest that subjective cognitive load, perceived task difficulty, success, and usability did not differ meaningfully between tablet and VR, addressing RQ1.

4 DISCUSSION, CONCLUSIONS, AND FUTURE WORK

Our preliminary results address both research questions. For RQ1, tablet and VR interfaces produced subjectively similar experiences, with no meaningful differences in reported cognitive load, perceived difficulty, success, or usability. For RQ2, although final customization outcomes were comparable across mediums, behavioral differences emerged: VR users took fewer actions and more time per action during customization. This pattern suggests that medium may shape how users explore and make decisions during customization, even when perceived experience and outcomes appear similar. While definitive conclusions cannot be drawn from this pilot, the large effect sizes in objective measures indicate that these trends may become more pronounced in a full study.

Future work will include path-based analyses (i.e., path length and variance in parameter space), examination of target difficulty effects, and administration of the Simulator Sickness Questionnaire. Several factors likely influenced results. The RPM interface is simple and affords limited customization; more complex interfaces may amplify medium-based differences. Target images were still being finalized, and variation in difficulty likely affected performance. Our sample was also highly familiar with customization interfaces, which may have reduced observed differences.

Overall, this pilot demonstrates that avatar customization can be operationalized as a cognitive-perceptual task. This methodological contribution lays the groundwork for studying how interface design choices shape performance, behavior, and experience in immersive environments.

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⁴<https://github.com/stylegan-human/StyleGAN-Human/blob/main/docs/Dataset.md>